Uncertainty: Error Bars, Confidence & Gradient

Workshop 3 — Module 7

Prof.Asoc. Endri Raco

2025/26

Data Visualization for Data Scientists
UNYT Tirana

Module Roadmap

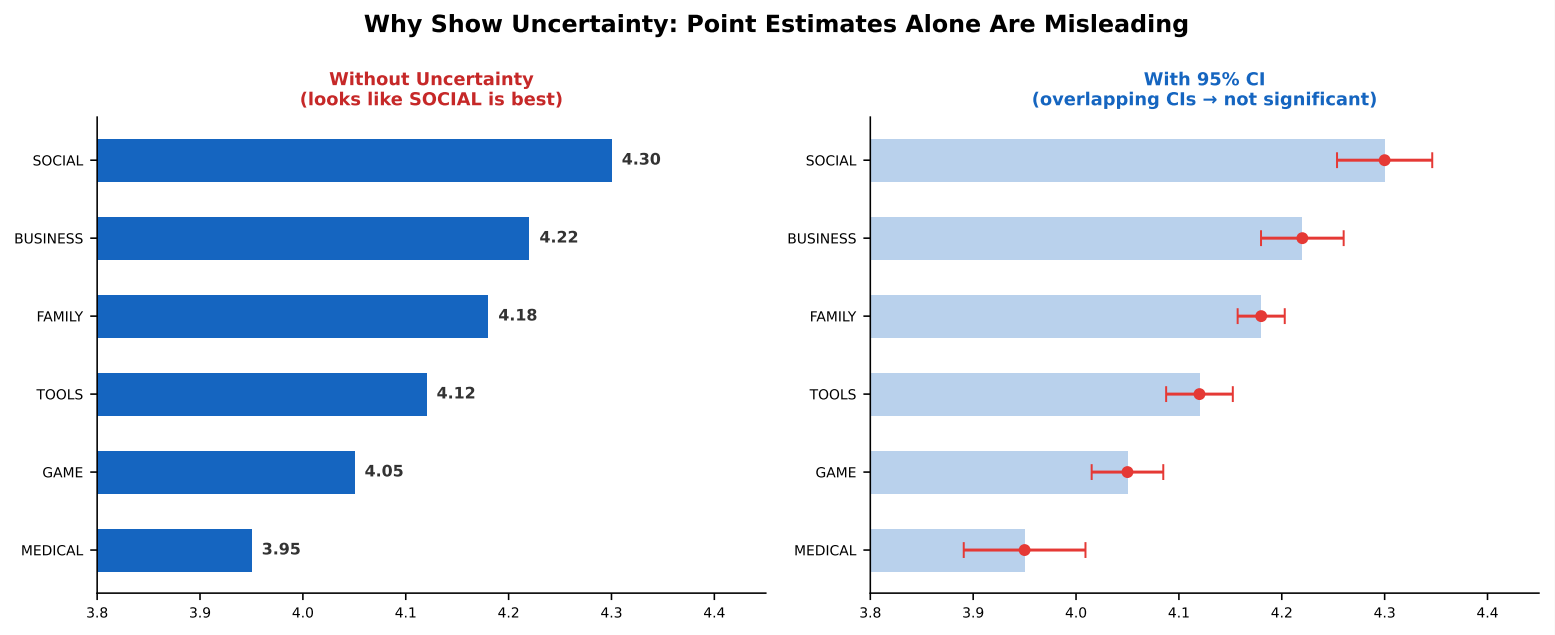
1. Why visualize uncertainty: point estimates are misleading
2. Three error bar types: SE, 95% CI, SD
3. Pointrange: the better error bar chart
4. Confidence ribbons: uncertainty on time series
5. Gradient density bands: continuous probability
6. Forest plots: meta-analysis style

Learning Outcome

You will always show uncertainty alongside point estimates, choose the correct error metric (SE vs CI vs SD), and produce publication-quality uncertainty visualizations in R and Python.

Why Show Uncertainty

Point Estimates Alone Are Misleading

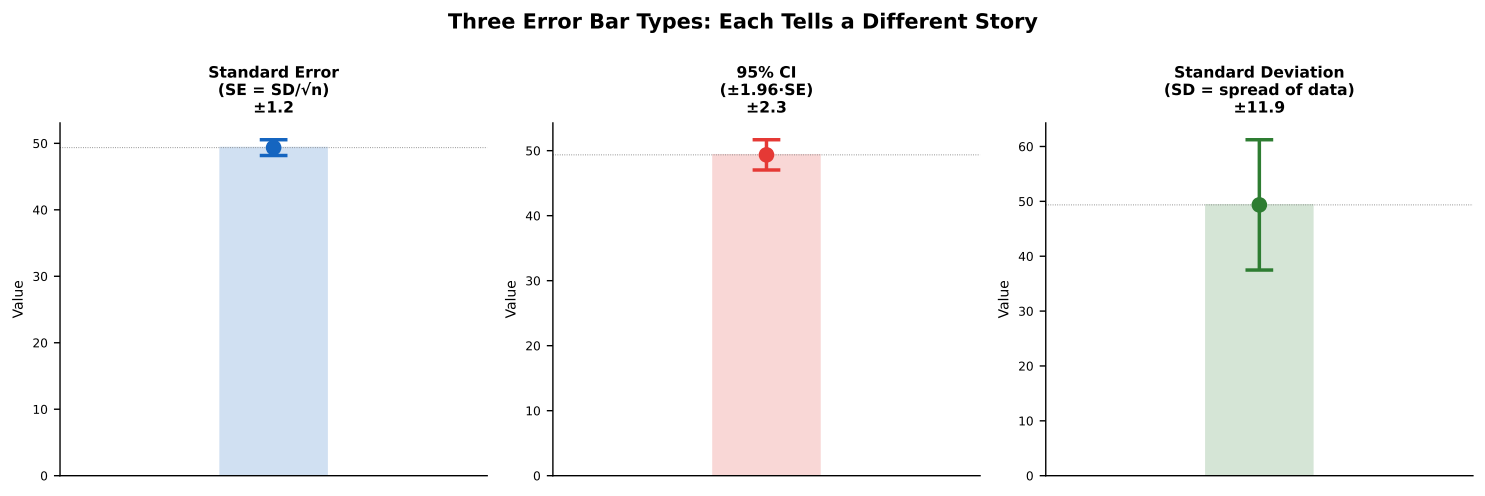


Pro-Tip

A bar chart of means without error bars implies precision that does not exist. Overlapping 95% CIs indicate the difference is likely not statistically significant. **Every mean needs an uncertainty interval.**

Error Bar Types

SE vs 95% CI vs SD



Data Insight

SE (standard error) = precision of the mean estimate. **95% CI** ($\pm 1.96 \cdot SE$) = range likely containing the true mean. **SD** = spread of individual observations. Use CI for inference; SD for describing variability. Always label which one you are showing.

Error Bars in R vs Python

R (ggplot2)

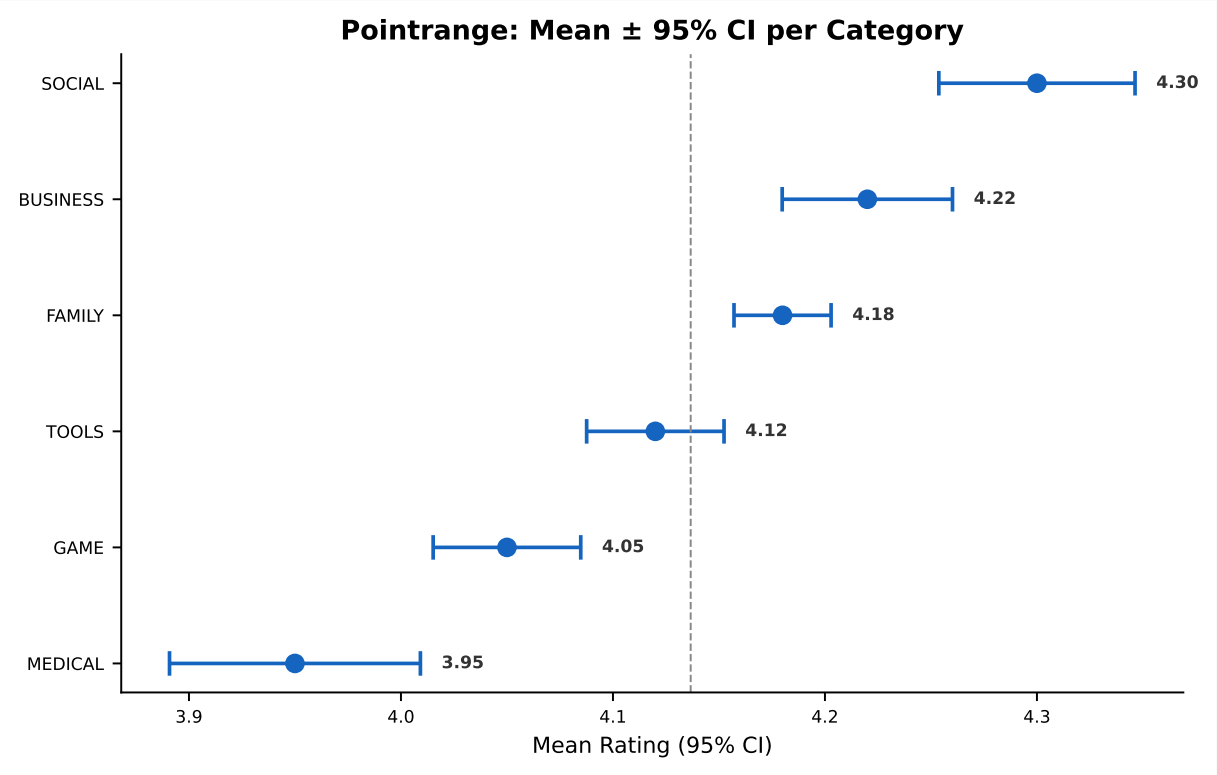
```r
# Auto-computed mean + 95% CI
ggplot(apps, aes(x = Category,
    y = Rating)) +
  stat_summary(
    fun.data = mean_cl_normal,
    geom = "errorbar",
    width = 0.2,
    color = "#E53935",
    linewidth = 1) +
  stat_summary(
    fun = mean,
    geom = "point",
    size = 3,
    color = "#1565C0") +
  coord_flip() +
  theme_minimal()

# mean_cl_normal   -> 95% CI
# mean_se          -> +/- 1 SE
# mean_sdl(mult=1)-> +/- 1 SD
```

Python

```python
from scipy.stats import sem

# Compute manually
grouped = apps.groupby("Category"
    )
stats = grouped["Rating"].agg(
    ["mean", "count", "std"])
stats["se"] = stats["std"] / \
    np.sqrt(stats["count"])
stats["ci95"] = 1.96 * stats["se"
    ]

# Plot
ax.errorbar(
    stats["mean"],
    range(len(stats)),
    xerr=stats["ci95"],
    fmt="o",
    color="#1565C0",
    markersize=6,
    capsize=5,
    linewidth=1.5)

# seaborn (auto CI):
```

Pointrange

Pointrange: The Better Bar Chart



Pro-Tip

Replace bar charts of means with **pointrange** plots: a point for the mean + horizontal lines for the CI. This eliminates the misleading bar area and forces the viewer to focus on the estimate and its precision.

Pointrange in Code

R

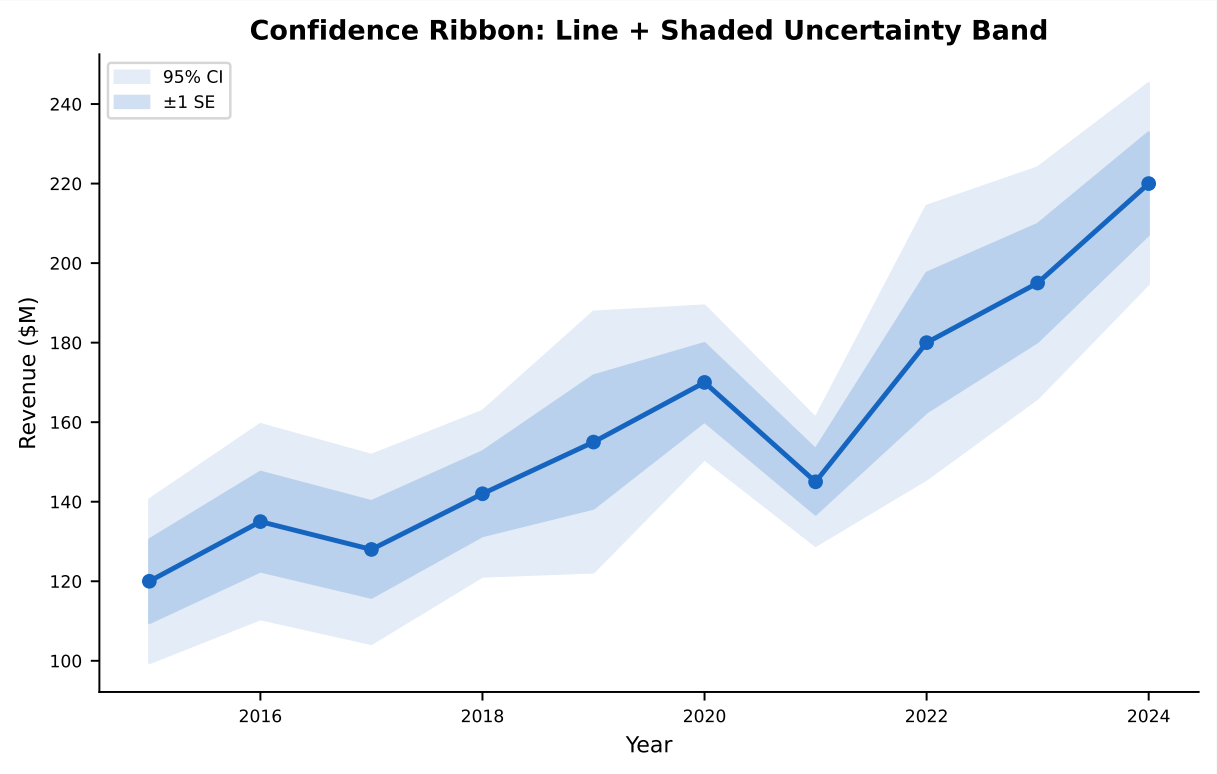
```r
ggplot(apps, aes(
    x = fct_reorder(Category,
      Rating, .fun = mean),
    y = Rating)) +
  stat_summary(
    fun.data = mean_cl_normal,
    geom = "pointrange",
    color = "#1565C0",
    size = 0.5,
    linewidth = 1) +
  coord_flip() +
  geom_hline(
    yintercept = mean(
      apps$Rating, na.rm = TRUE),
    linetype = "dashed",
    color = "grey50") +
  theme_minimal() +
  labs(title = "Mean Rating (95%
      CI)",
       x = NULL, y = "Rating")
```

Python

```python
fig, ax = plt.subplots()

# Sorted by mean
stats_sorted = stats.sort_values(
    "mean")

ax.errorbar(
    stats_sorted["mean"],
    range(len(stats_sorted)),
    xerr=stats_sorted["ci95"],
    fmt="o",
    color="#1565C0",
    markersize=7,
    capsize=5,
    linewidth=1.5,
    capthick=1.5)

ax.axvline(
    x=stats_sorted["mean"].mean()
        ,
    color="grey", lw=0.8, ls="--"
        )

ax.set_yticks(
```

Confidence Ribbons

Ribbon: Uncertainty on Time Series



Data Insight

A ribbon shades the region between the lower and upper CI bounds around a trend line. Use a lighter shade for the wider 95% CI and a darker shade for the narrower ±1 SE. ggplot2's `geom_smooth(se = TRUE)` does this automatically; matplotlib requires `ax.fill_between()`.

Ribbon in R vs Python

R

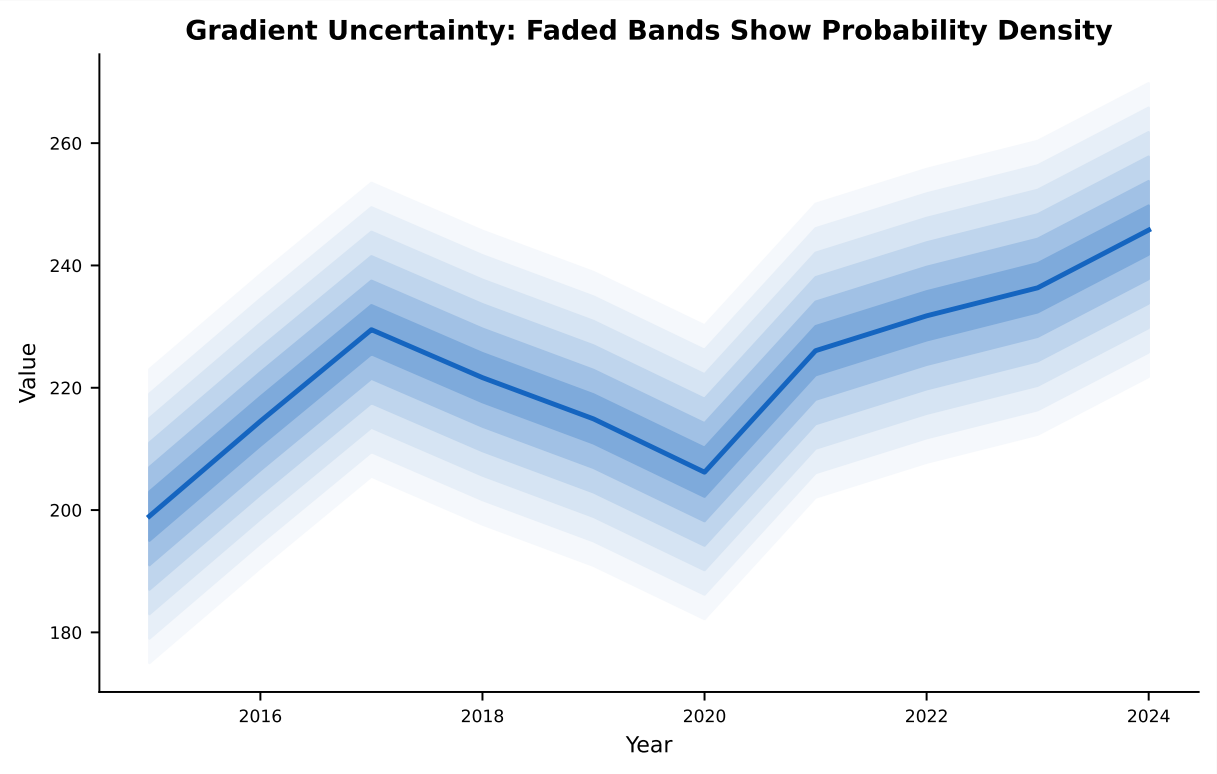
```r
# Automatic via geom_smooth
ggplot(df, aes(x = year,
               y = revenue)) +
  geom_smooth(method = "lm",
    se = TRUE,          # CI band
    color = "#1565C0",
    fill = "#1565C0",
    alpha = 0.15) +
  geom_point(color = "#1565C0") +
  theme_minimal()

# Manual ribbon
ggplot(df, aes(x = year)) +
  geom_ribbon(aes(
    ymin = lower_ci,
    ymax = upper_ci),
    fill = "#1565C0",
    alpha = 0.15) +
  geom_line(aes(y = mean),
    color = "#1565C0", lw = 1.2)
```

Python

```python
# Manual ribbon
ax.plot(years, means, "-o",
    color="#1565C0", lw=2)

# 95% CI band
ax.fill_between(years,
    means - 1.96 * ses,
    means + 1.96 * ses,
    alpha=0.12, color="#1565C0",
    label="95% CI")

# +/- 1 SE (darker)
ax.fill_between(years,
    means - ses,
    means + ses,
    alpha=0.25, color="#1565C0",
    label="1 SE")

ax.legend(fontsize=7)
```

Gradient Density

Gradient Bands: Continuous Probability

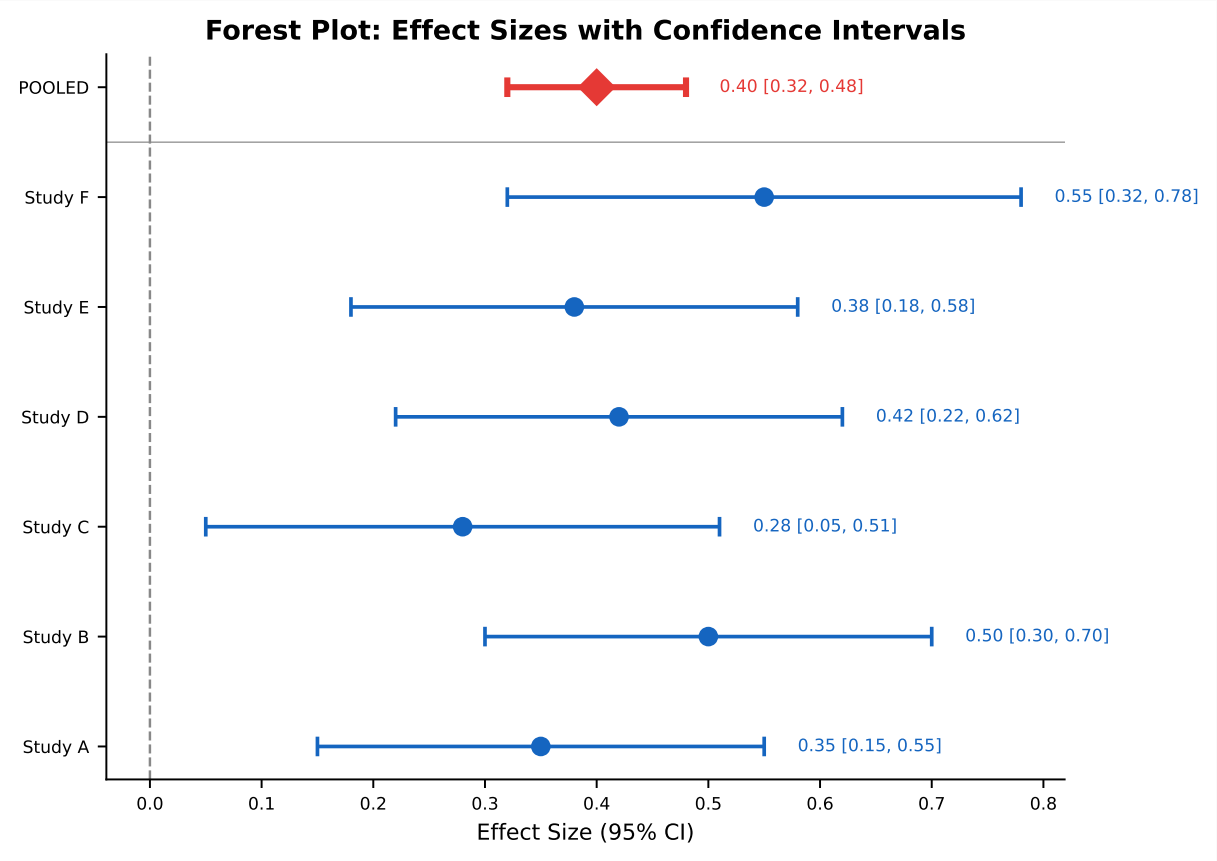


Pro-Tip

Gradient (faded) bands convey continuous probability: darker = more likely, lighter = less likely. This avoids the sharp boundary problem of single-CI ribbons. In R: `ggdist::stat_lineribbon()`. In Python: loop `fill_between()` with decreasing alpha at increasing multiples of SE.

Forest Plots

Forest Plot: Meta-Analysis Style



Data Insight

Forest plots show effect sizes with CIs for multiple studies, with a pooled (diamond) estimate at the bottom. The vertical dashed line at zero (or the null value) is the reference. If a CI crosses zero, the effect is not significant. Standard in medical and social science meta-analysis.

Chart Selection

Uncertainty Visualization Comparison

Uncertainty Visualization Comparison

Chart	Shows	Use For	R	Python
Error bar	Mean ± SE/CI/SD	Categorical comparison	`geom_errorbar / geom_pointrange`	`ax.errorbar(capsize=)`
Pointrange	Point + range	Cleaner than errorbar	`geom_pointrange()`	`ax.errorbar(fmt='o')`
Ribbon	Shaded band around a line	Time series uncertainty	`geom_ribbon() / geom_smooth(se=T)`	`ax.fill_between()`
Gradient	Faded nested bands	Continuous probability	`ggdist / distributional`	`Manual alpha loop`
Forest plot	Effect + CI per study	Meta-analysis	`forestplot / manual`	`Manual errorbar loop`

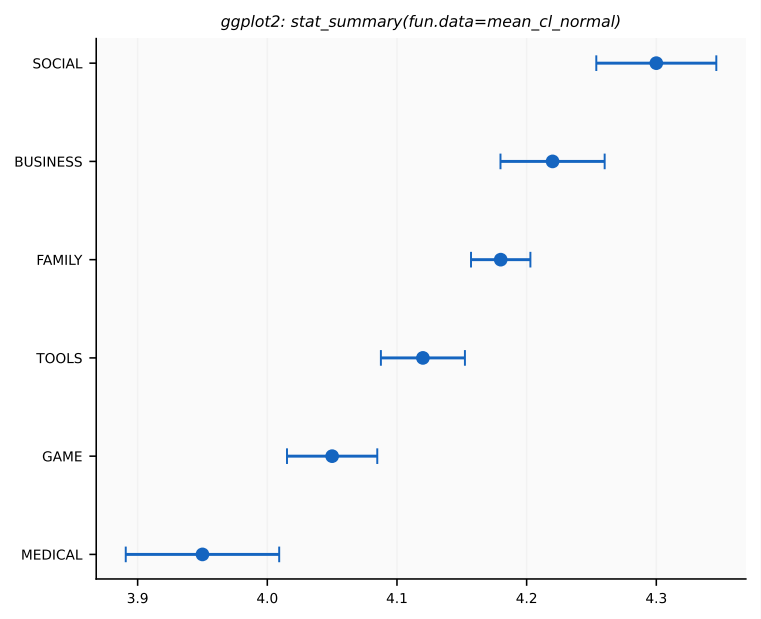
When to Use Which

Scenario	Chart	Error Metric
Compare group means	Pointrange	95% CI
Time series trend	Ribbon / gradient	95% CI or SE
Describe data spread	Error bar with SD	SD
Meta-analysis	Forest plot	95% CI per study
Regression fit	`geom_smooth(se=T)`	CI on fitted line
Forecast	Gradient / fan chart	Prediction intervals

Complete Examples

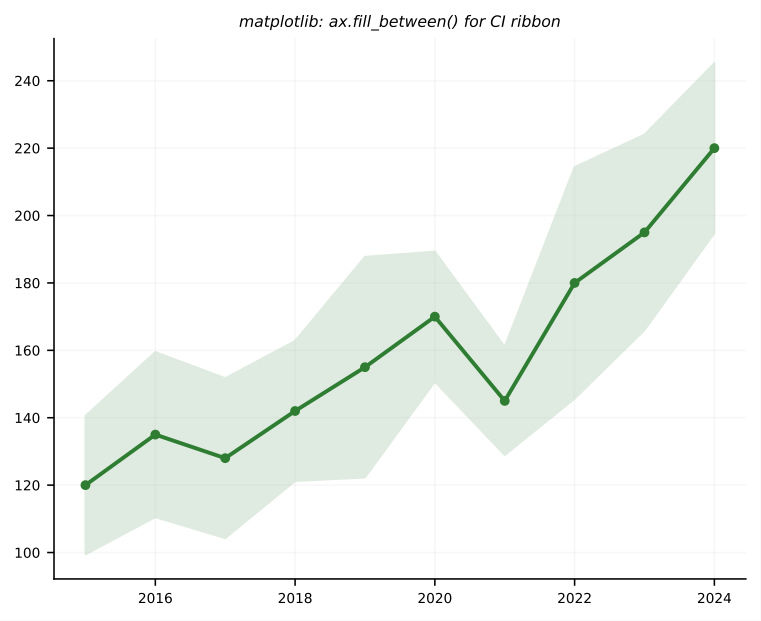
R: Pointrange (Publication Quality)

```r
apps |>
  filter(!is.na(Rating),
    Category %in% top6) |>
  ggplot(aes(
    x = fct_reorder(Category,
      Rating, .fun = mean),
    y = Rating)) +
  stat_summary(
    fun.data = mean_cl_normal,
    geom = "pointrange",
    color = "#1565C0",
    size = 0.5,
    linewidth = 1) +
  coord_flip() +
  geom_hline(
    yintercept = grand_mean,
    linetype = "dashed",
    color = "grey50") +
  theme_minimal() +
  labs(
    title = "SOCIAL Has the
      Highest Mean Rating",
    subtitle = "Mean +/- 95% CI",
    x = NULL, y = "Rating")
```



Python: Confidence Ribbon

```python
fig, ax = plt.subplots(
    figsize=(7, 5))

# Trend line
ax.plot(years, revenue, "-o",
    color="#2E7D32", lw=2,
    markersize=5)

# 95% CI ribbon
ax.fill_between(years,
    revenue - 1.96 * se,
    revenue + 1.96 * se,
    alpha=0.15, color="#2E7D32")

ax.set_title(
    "Revenue Trend with 95% CI",
    fontweight="bold")
ax.set_xlabel("Year")
ax.set_ylabel("Revenue ($M)")
ax.spines["top"].set_visible(False)
ax.spines["right"].set_visible(
     False)
```



Synthesis

Key Takeaways

1. **Never show a mean without uncertainty**. Bar charts of means without error bars are actively misleading.
2. **SE** = precision of the mean; **95% CI** = range containing the true mean; **SD** = data spread. Always label which one you show.
3. **Pointrange** replaces the bar+errorbar pattern: `stat_summary(geom = "pointrange")` (R) or `ax.errorbar(fmt="o")` (Python).
4. **Ribbons** show uncertainty on time series: `geom_ribbon()` or `ax.fill_between()`.
5. **Gradient bands** avoid sharp CI boundaries by fading with probability distance.
6. **Forest plots** summarize multiple studies with a pooled estimate — standard in meta-analysis.

Essential Readings

- Cumming, G. (2014). "The new statistics: Why and how." *Psychological Science*, 25(1), 7–29.
- Kay, M. et al. (2016). "When (ish) is my bus?" *CHI '16*. (Gradient uncertainty.)
- Wilke, C. O. (2019). *Fundamentals of Data Visualization*, Chapter 16 (Visualizing Uncertainty).
- Wickham, H. (2016). *ggplot2*, Chapter 5.4 (Statistical Summaries).

Next Module: Small Multiples & Faceted Displays (W03-M08)